

Hyponatremia — the 3-Lab Diagnostic Approach

Serum osm "is it real?" → urine osm "is ADH on?" → urine Na "are the kidneys seeing enough fluid?" · skip the unreliable bedside volume guess · diagnosis only — treatment is a separate pathway

1 · Serum osm — is it real?

SOsm	Read	Causes / action
<290	True hypo-osmolar hypoNa	→ Step 2
Normal	Pseudohyponatremia	↑TG, ↑protein (myeloma) dilute whole plasma → send whole-blood (direct ISE) Na
>290	Hyperosmolar	Active osmoles (glucose, mannitol, sorbitol) pull water from cells; inactive (BUN, EtOH) raise osm but do NOT cause hypoNa

Corrected Na = measured + $1.6 \times (\text{glucose} - 100)/100$ — glc 500 hides ~6.5 mEq/L

2 · Urine osm — is ADH on or off?

UOsm	Read	Differential
<100	ADH OFF (kidney maximally diluting — intake is the problem)	Primary polydipsia · beer potomania · tea-and-toast (low solute)
>100	ADH ON	Everything else → Step 3: is the ADH appropriate?

- ADH (pituitary) fires on low volume/ECV (baroreceptors) or high SOsm → collecting duct reclaims water → concentrated urine

3 · Urine Na — do the kidneys see enough fluid?

UNa	Physiology	Differential
<25	RAAS on → Na reclaimed; ADH <i>appropriate</i>	True hypovolemia (GI loss, bleeding) · low ECV w/ high total volume: HF, cirrhosis
>40	Kidneys perfused → ADH <i>inappropriate</i>	SIADH · hypothyroidism · adrenal insufficiency · diuretics (exception: salt wasting falsely ↑UNa)

- **SIADH triggers:** antidepressants/antiepileptics/antipsychotics, surgery, pain, CNS injury, lung disease, malignancy — patient euvolemic
- **Cerebral salt wasting** (esp. SAH): SIADH-pattern labs but HYPOVOLEMIC
- Always check **TSH + cortisol** before calling SIADH — replacement corrects both impostors

Worked cases (Na 125 in all three)

Labs	Diagnosis
SOsm 270 / UOsm 60 / UNa 65 — drinker	Beer potomania (ADH off, low solute)
SOsm 310 / UOsm 800 / UNa 20 — glc 500, DKA	Hyperosmolar hypoNa (corrected ~131) + hypovolemia
SOsm 265 / UOsm 400 / UNa <10 — vomiting + diarrhea	True hypovolemic hypoNa (appropriate ADH)

Do / AVOID

- **DO** confirm hypo-osmolality before any further workup — artifact and glucose first
- **DO** read UOsm before UNa: a dilute urine ends the workup (intake problem)
- **DO** check TSH + AM cortisol before diagnosing SIADH
- **AVOID** interpreting urine Na on diuretics — salt wasting voids the rule
- **AVOID** leading with a bedside volume exam — labs first, exam to break SIADH-vs-CSW ties
- **AVOID** rapid correction of chronic hypoNa (ODS risk) — treatment is its own pathway

Framework unchanged in current guidance — 2024 Austrian Society for Nephrology consensus uses the same SOsm/UOsm/UNa triad.